

DIGI XBEE³ ZIGBEE 3.0

Easy-to-add connectivity in a compact, low-power, low-profile footprint.

Digi XBee3[®] modules accelerate time to market for designers, OEMs and solution providers by quickly enabling wireless connectivity and easy-to-add functionality. Building on industry-leading technology, pre-certified Digi XBee3[™] modules offer the flexibility to switch between multiple frequencies and wireless protocols as needed.

Digi XBee3 ZigBee 3.0 offers a fully interoperable ecosystem covering all vertical markets including building automation, smart energy, digital health, intelligent lighting, and others.

With Digi Remote Manager[®], Digi XBee3 modules can be easily configured and controlled from a simple, central platform. Built-in Digi TrustFence[®] security, identity and data privacy

features use more than 175 controls to protect against new and evolving cyber threats. MicroPython and XCTU software tools simplify adding functionality, configuration and testing.

From edge computing to future migration, Digi XBee modules offer size, weight, power and performance advantages ideal for scalable device connectivity. A versatile addition to the expanding Digi XBee Ecosystem of wireless modules, adapters and software, the Digi XBee3 Series is engineered to accelerate development and deployment.

SIZE AND FLEXIBILITY

- At 13 mm x 19 mm, the new Digi XBee3 micro form factor allows for more compact and portable applications
- Digi XBee3 is one module for all protocols including: ZigBee, 802.15.4, DigiMesh and BLE, all configurable via Digi XCTU

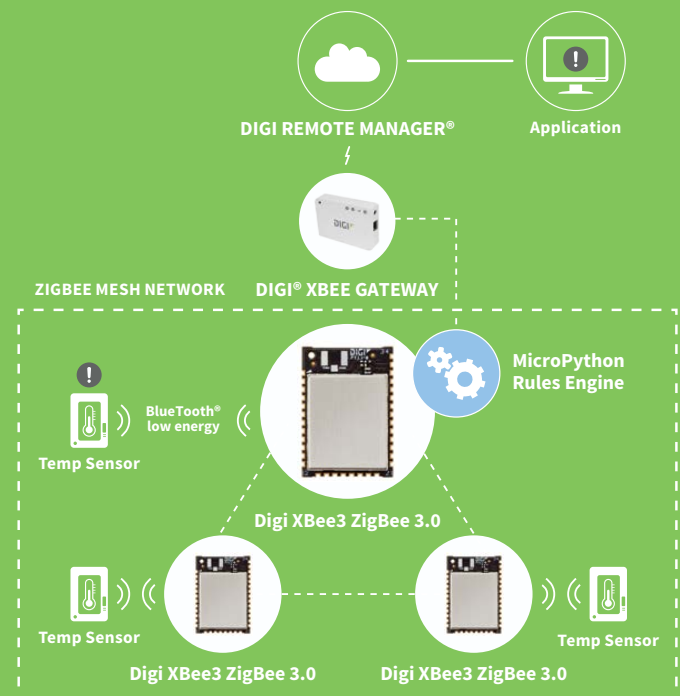
PROGRAMMABILITY

- Eliminate the need for an external microcontroller and create smart end nodes using MicroPython
- Dual-mode radio for local configuration over Bluetooth[®] low energy using the Digi XBee Mobile app

SECURITY

- Intrinsic IoT security with Digi TrustFence[®], a layered approach securing the edge device, through the gateway, into and out of the IoT

APPLICATION EXAMPLE



RELATED PRODUCTS AND SERVICES



Development Kits



Digi XCTU



Digi TrustFence[®]



Digi Remote Manager[®]



Digi Design Services

SPECIFICATIONS	Digi XBee3 ZigBee 3.0		Digi XBee3 PRO ZigBee 3.0	
PERFORMANCE				
TRANSCEIVER CHIPSET	Silicon Labs EFR32MG SoC			
DATA RATE	RF 250 Kbps, Serial up to 1 Mbps			
INDOOR/URBAN RANGE*	Up to 200 ft (60 m)	Up to 300 ft (90 m)		
OUTDOOR/RF LINE-OF-SIGHT RANGE*	Up to 4000 ft (1200 m)	Up to 2 miles (3200 m)		
TRANSMIT POWER	+8 dBm	+19 dBm		
RECEIVER SENSITIVITY (1% PER)	-103 dBm Normal Mode			
FEATURES				
SERIAL DATA INTERFACE	UART, SPI, I²C			
CONFIGURATION METHOD	API or AT commands, local or over-the-air (OTA)			
FREQUENCY BAND	ISM 2.4 GHz			
FORM FACTOR	Micro, Through-Hole, Surface Mount			
INTERFERENCE IMMUNITY	DSSS (Direct Sequence Spread Spectrum)			
ADC INPUTS	(4) 10-bit ADC inputs			
DIGITAL I/O	15			
ANTENNA OPTIONS	Through-Hole: PCB Antenna, U.FL Connector, RPSMA Connector SMT: RF Pad, PCB Antenna, or U.FL Connector Micro: U.FL Antenna, RF Pad, Chip Antenna			
OPERATING TEMPERATURE	-40° C to +85° C			
DIMENSIONS (L X W X H)	Through-Hole: 0.960 x 1.087 in (2.438 x 2.761 cm) SMT: 0.866 x 1.33 x 0.120 in (2.199 x 3.4 x 0.305 cm) Micro: 0.533 x 0.76 x 0.087 in (13 x 19 x 2 mm)			
PROGRAMMABILITY				
MEMORY	1 MB / 128 KB RAM			
CPU/CLOCK SPEED	HCS08 / up to 50.33 MHz			
NETWORKING AND SECURITY				
PROTOCOL	ZigBee® 3.0			
ENCRYPTION	128/256 bit AES			
RELIABLE PACKET DELIVERY	Retries/Acknowledgements			
IDS	PAN ID and addresses, cluster IDs and endpoints (optional)			
CHANNELS	16 channels			
POWER REQUIREMENTS				
SUPPLY VOLTAGE	2.1 to 3.6V			
TRANSMIT CURRENT	40 mA @ 8 dBm	135 mA @ 19 dBm		
RECEIVE CURRENT	15 mA			
POWER-DOWN CURRENT	1.7 micro Amp @ 25 degrees C			
REGULATORY APPROVALS				
FCC, IC (NORTH AMERICA)	Yes	Yes		
ETSI (EUROPE)	Yes	No		

*Range figure estimates are based on free-air terrain with limited sources of interference. Actual range will vary based on transmitting power, orientation of transmitter and receiver, height of transmitting antenna, height of receiving antenna, weather conditions, interference sources in the area, and terrain between receiver and transmitter, including indoor and outdoor structures such as walls, trees, buildings, hills, and mountains.

Mouser Electronics

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